

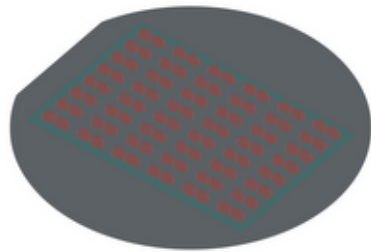
Post-mold device replication and assembly

The same hybrid mold supports repeated plate-format PDMS casting and biological assembly.

Casting and curing

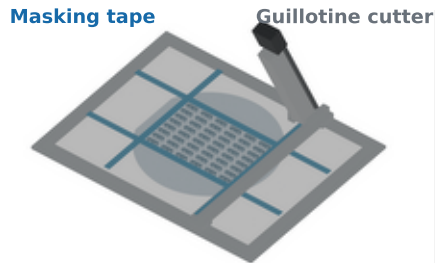
Release and trimming

Assembly and preparation



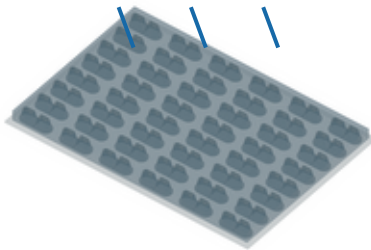
1 Hybrid mold ready

Parylene-coated wafer with bonded resin well inserts



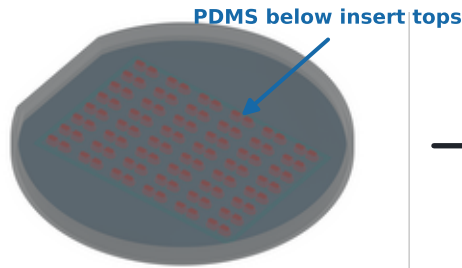
6 Tape and trim

Tape inside the wall marks, cut, and chamfer the corners



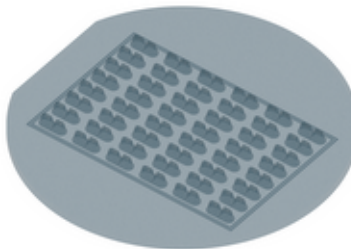
7 Bond and autoclave

Plasma-bond to glass, thermally stabilize, and autoclave at 121 degrees C for 15 min



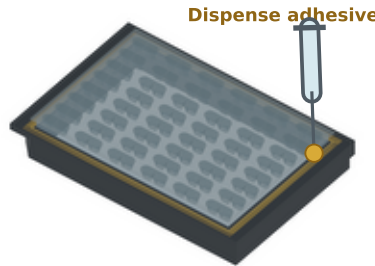
2 Pour PDMS

Cover the full wafer, stopping below the insert tops



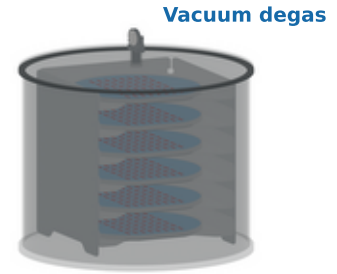
5 Demold cast

Release the circular PDMS cast with wells facing upward



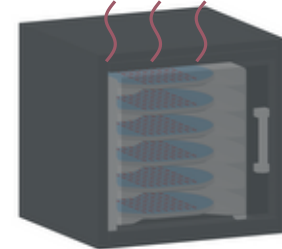
8 Frame assembly

Fill the inverted frame groove and seat the glass-side-up device



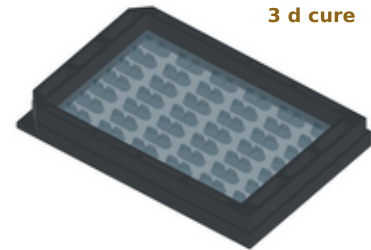
3 Degas in rack

Load filled molds into the six-shelf rack and degas
100 degrees C, 1 h



4 Cure in rack

Transfer the loaded rack directly to the curing oven



9 Cure and prepare

After the 3 d adhesive cure, turn the framed device upright and plasma-treat